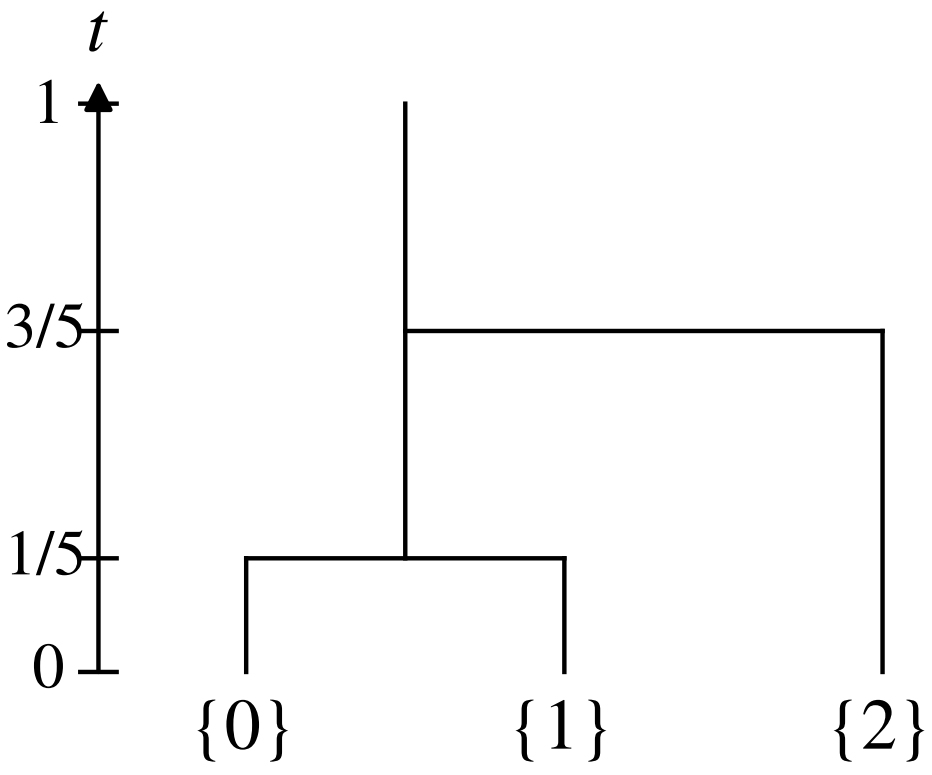

$$w_{\{0\}} = w_{\{1\}} = 1/5$$
$$w_{\{2\}} = 3/5$$
$$w_{\{0,1\}} = 2/5$$
$$w_{\{0,1,2\}} = 2/5$$